THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — SCIENCE · MEDIUM LEVEL CHAPTER TEST SET 1

Chapter 9: **Light - Reflection and Refraction** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. An object is placed at the centre of curvature of a concave mirror of focal length 12 cm. The distance of the image from the mirror is 1

(a) 12 cm **(b)** 36 cm **(c)** 24 cm **(d)** 6 cm

2. If n = sin i / sin r for glass is 1.5 and sin i = 0.75, then sin r is 1

(a) 1.125 **(b)** 0.75 **(c)** 0.25 **(d)** 0.5

3. An object is placed 40 cm from a convex lens of f = 20 cm. The image distance is 1

(a) 10 cm **(b)** 20 cm **(c)** 60 cm **(d)** 40 cm

4. A concave mirror of f = 15 cm forms an image of the same size as the object. The object distance is 1

(a) 30 cm **(b)** 45 cm **(c)** 15 cm **(d)** 7.5 cm

5. Speed of light in a medium of n = 2 is 1

(a) 6×10^8 m/s **(b)** 1.5×10^8 m/s **(c)** 2×10^8 m/s **(d)** 3×10^8 m/s

6. An object 3 cm high is at 20 cm from a convex lens of f = 10 cm. The image height is 1

(a) 3 cm (erect) **(b)** 6 cm **(c)** 3 cm (inverted) **(d)** 1.5 cm

7. An object is placed 18 cm from a concave mirror of f = 12 cm. The image distance is 1

(a) 24 cm **(b)** 36 cm **(c)** 12 cm **(d)** 18 cm

8. Light enters from air into a medium at an angle of 45 degrees and refracts at 30 degrees. The refractive index is 1

(a) $\sqrt{2}$ **(b)** 2 **(c)** $\sqrt{3}$ **(d)** $1/\sqrt{2}$

9. A convex lens of f = 15 cm forms a real image at 60 cm. The object distance is 1

(a) 45 cm **(b)** 30 cm **(c)** 20 cm **(d)** 15 cm

10. A convex mirror has f = 20 cm. An object is at 30 cm. The image distance is 1

(a) 12 cm behind the mirror **(b)** 60 cm behind the mirror

(c) 20 cm behind **(d)** 12 cm in front

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. An object is placed 30 cm from a concave mirror. The image is formed at 30 cm on the same side. Find the focal length. 2

12. The refractive index of a medium is 1.6. Find the speed of light in that medium. 2

13. Why can a concave lens never form a real image? 2

14. Find the magnification when a convex mirror of f = 15 cm forms an image at 7.5 cm behind it of an object at 15 cm. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. An object 6 cm high is placed 20 cm in front of a concave mirror of f = 15 cm. Find the image distance, size and nature. 3

16. A concave lens of f = -10 cm forms an image of an object at 20 cm. Find the image distance, nature and magnification. 3

17. Differentiate between the images formed by a concave mirror and a convex lens for an object at 2F. 3

OR

A convex mirror has a focal length of 20 cm. An object is placed 20 cm in front of it. Find the image distance and magnification. 3

18. A 4 cm high object is placed 12 cm from a convex lens of f = 8 cm. Find the image height and nature. 3

OR

State what will happen to the focal length of a lens if it is dipped in water. Explain. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. An object 3 cm high is placed 20 cm from a concave mirror of focal length 12 cm. Find the position, nature, size of the image and magnification. Draw the ray-diagram rules used. 5

OR

The refractive indices of kerosene, water, glass and diamond are 1.44, 1.33, 1.5 and 2.42. 5
(a) Arrange in increasing order of optical density
(b) find the speed of light in diamond
(c) state in which light bends the most.

20. A convex mirror has a radius of curvature of 40 cm. An object 5 cm tall is 20 cm from it. Find the image distance, size and nature. 5

OR

An object is placed at 15 cm from a lens whose focal length is 10 cm. 5
(a) Find the image distance
(b) the magnification
(c) the nature of the image. Then repeat if the lens is concave with f = -10 cm.

END OF QUESTION PAPER — Total: 40 Marks